

```
import cv2
import matplotlib.pyplot as plt

# Read grayscale image
image = cv2.imread("medical_image.jpg", cv2.IMREAD_GRAYSCALE)

if image is None:
    print("Image not found!")
    exit()

# -----
# Simulate Low Pixel Resolution
# -----
small = cv2.resize(image, (64, 64), interpolation=cv2.INTER_LINEAR)
low_pixel = cv2.resize(small,
                        (image.shape[1], image.shape[0]),
                        interpolation=cv2.INTER_NEAREST)

# -----
# Simulate Low Intensity Resolution (4-bit)
# -----
low_intensity = (image // 16) * 16

# -----
# Display Results
# -----
plt.figure(figsize=(12,4))

plt.subplot(1,3,1)
plt.imshow(image, cmap='gray')
plt.title("Original Image")
plt.axis("off")

plt.subplot(1,3,2)
plt.imshow(low_pixel, cmap='gray')
plt.title("Low Pixel Resolution")
plt.axis("off")
```

```
plt.subplot(1,3,3)
plt.imshow(low_intensity, cmap='gray')
plt.title("Low Intensity Resolution")
plt.axis("off")

plt.tight_layout()
plt.show()
```

Image not found!

```
-----
error                                Traceback (most recent call last)
/tmp/ipykernel_6367/1351048503.py in <cell line: 0>()
    12 # Simulate Low Pixel Resolution
    13 # -----
--> 14 small = cv2.resize(image, (64, 64), interpolation=cv2.INTER_LINEAR)
    15 low_pixel = cv2.resize(small,
    16                        (image.shape[1], image.shape[0]),

error: OpenCV(4.13.0) /io/opencv/modules/imgproc/src/resize.cpp:4208: error: (-215:Assertion failed)
!ssize.empty() in function 'resize'
```

Next steps: [Explain error](#)

